

CHAPTER

20



Pandu

Learning Objectives

- ⇒ To learn the concept of Pandu
- ⇒ To understand the Nidana Panchaka of Pandu mentioned in Ayurveda literatures
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease
- ⇒ To learn the Pratyatma Lakshana (cardinal feature) of Pandu
- ⇒ To know about the Purvarupa of Pandu with current understanding
- ⇒ To know about the Upadrava of Pandu with current understanding
- ⇒ To understand the Sadhyasadyata of Pandu with current understanding
- ⇒ To know about the 5 types of Pandu
- ⇒ To know the concept of Dosha dominant Pandu

Introduction

पाण्डुना वक्ष्यमाणहरितादिवर्णेभ्यः प्रधानेन वर्णेनोपलक्षितो रोगः
पाण्डुरोगः।

उतं हि सुश्रुते 'सर्वेषु चैतेषु हि पाण्डुभावो यतोऽधिकोऽतः खलु
पाण्डुरोगः' (सु. उ. अ. 44) इति। Chakra, Ca.Ci. – 16/1-2

The disease *Pandu* is considered to be related to discolouration of the body. The discolouration of the body may range from yellowish white discolouration to greenish discolouration of the body. *Ayurveda* literatures say the discolouration of the skin resembles colour of *Ketaki Dhuli* (see Figure 58) (particles of *Ketaki*) which is yellowish white discolouration. As per *Ayurveda* understanding, the term *Varna* refers not only discolouration, abnormal texture of the skin should also be considered. This is the disease involving *Rasavaha Srotas* and is a *Pitta* dominant disease with *Sanga* and *Vimargagamana* are the nature of *Srotodushti* happening during the course of the illness.

Pratyatma Lakshana of Pandu: The following are considered as *Pratyatma Lakshana* (cardinal features) of the disease *Pandu*.

सोऽल्परकोऽल्पमेदस्को निःसारः शिथिलेन्द्रियः।
वैवर्ण्यं भजते, तस्य हेतुं शृणु सलक्षणम्॥6॥

Ca.Ci. – 16/6



Figure 58: Colour of Ketaki Dhuli (yellowish white) of Pandu Roga

- **Alparakta**¹ – less volume of the blood
- **Alpa Medas**² – less amount of fat in the body
- **Nissara**³ – poor nutritional status of the body
- **Shithilendriya**⁴ – poor functioning of the *Indriya* either *Jnanendriya* or *Karmendriya*
- **Vaivarnya**⁵ – discolouration of the body or abnormal texture of the skin

Samanya Nidana of Pandu

क्षाराम्ललवणात्युष्णविरुद्धासात्म्यभोजनात्।
निष्पावमापिण्याकतिलतैलनिषेवणात्॥७॥
विदग्धेऽन्त्रे दिवाखनाद्यायामानैथुनात्तथा।

प्रतिकर्मतुर्वैषम्याद्वेगानां च विधारणात्॥८॥
कामचिन्ताभयक्रोधशोकोपहतचेतसः।
व्यायमम्लं लवणानि मद्यं मृदं दिवास्वप्नतीव तीक्ष्णम्।
निषेवमाणस्य विदूष्य रक्तं कुर्वन्ति दोषास्त्वचि पाण्डुभावम्॥९॥

Ca.Ci. – 16/7-9

Su.Ut. – 44/3

- **Kshara**⁶ – excess consumption of sour substances
- **Amla**⁷ – excess consumption of salty substances
- **Lavana**⁸ – excess consumption of substances having
- **Atyushna**⁹ – excess consumption of substances having
- **Ushna Virya** (hot potency)
- **Viruddha**¹⁰ – excess consumption of *Viruddha* substances
- **Asatmya Bhojana**¹¹ – excess consumption of unwholesome substances

G K

- 1 This is attributable to the decreased amount of blood in the body either due to the blood loss or due to poor formation of blood cells in aplastic anemia or due to the deficiency of nutrients necessary for RBC formation or due to haemolysis in the body.
 - 2 As the person is having less amount of blood in the body reaching the every cell of the body, that refers to the mobilisation of fat situated in the different parts of the body, such as buttocks, breasts, abdomen or subcutaneous tissue of the body. Because of fat utilisation, the person will have less amount of stored fat in the body.
 - 3 This is attributable to the poor absorption of nutrients either deficiency of proteins or deficiency of intrinsic factor or deficiency of RBC forming substances or frequent infection in any part of the body.
 - 4 This is attributable to poor nutrition because of less volume of blood as there will be shift of the blood circulation from the periphery to the core of the body such as brain and heart. Poor blood supply to the skin or ear or tongue or eyes or nose or Karmendriya such as hands, feet, vocal cord, anus and urethra or vagina leading to poor functioning of these *Indriya*.
 - 5 **वर्णभेदेन ग्लानिहर्षरीक्ष्यस्नेहा व्याख्याताः॥**
This is attributable to abnormal discolouration either cyanosis or pallor or it can be abnormal texture such as dryness or cracks of the skin. Cracks in the skin can occur due to peripheral vaso constriction.
 - 6 Mucosa of the stomach has three types of cells – mucous cells, peptic or chief cells and oxyntic or parietal cells. Mucous cells are responsible for the production of mucus; peptic cells are producing the pepsinogen and oxyntic cells secrete hydrochloric acid and intrinsic factor. When an individual has the habit of consuming more amount of alkaline substances frequently, that will lead to the neutralisation of hydrochloric acid. This will lead to destruction of the oxyntic cells in due course of time. This will lead to the hampered secretion of intrinsic factor which is an essential factor for the absorption of vitamin B12. Thus, when an individual has the habit of continuous consumption of alkaline substances, he might be the sufferer of pernicious anemia.
 - 7 The term Amla refers to sour taste. If an individual consumes more amount of Amla substances, that will be responsible for the increased hydrogen ion concentration in the stomach. This may be responsible for the initiation of inflammation of alimentary canal either gastritis or enteritis. The mechanism is responsible for the increased peristaltic movements and the final outcome is diarrhoea. Because of diarrhoea, there will be poor absorption of iron, vitamin B12 etc from small intestine. This will lead to the manifestation of anemia.
 - 8 The term Lavana refers to salt taste. If an individual consumes more amount of Lavana substances, that will be responsible for the increased sodium ion concentration in the stomach. Normally, 20-30 gms of sodium are secreted in the intestinal secretions each day and 5-8 gms of sodium is consumed by an individual. To prevent the net loss of sodium from faecal matter, intestines will absorb 25 to 35 gms of sodium each day. This amount is equal to 1/7th of all the sodium present in the body.
 - Due to the excess salt consumption, the water is getting transported into the alimentary canal via lumen through the process of diffusion. This will lead to the increase of content of alimentary canal.
 - When an individual consumes excess salt, the person will consume excess amount of water. This also increase the volume of the content in the alimentary canal.
 - Both of the above mechanisms may lead to excess peristaltic movements, thereby the absorption of nutrients such as vitamin B12, folic acid, intrinsic factor which are necessary for formation of blood will hamper.
 - 9 The term Atyushna refers to the substances having hot potency such as chilli etc. This will lead to the inflammation of alimentary canal leading to the hypermotility and thus leading to the malabsorption syndrome.
 - 10 When an individual is consuming Viruddha substances in the form of foods and drinks may lead to the affliction of Rakta Dhatu which is liquid in nature, thereby may lead to the affliction of Rasa and Rakta Dhatu because Rasa and Rakta Dhatu are liquid in nature. This may show the affliction of skin disease.
- पाण्ड्यान्धवीसपेदकोदराणां विस्फोटकोन्मादभगन्दराणां। मूर्च्छामदाध्मानगलग्रहाणां पाण्डामयस्यामविषस्य चैव॥१०२॥
- शोधांम्लपित्तज्वरपीनसानाम्। सन्तानदोषस्य तथैव मृत्यो- विरुद्धमन्त्रं प्रवदन्ति हेतुम्॥१०३॥
- Pandu is also manifested due to the continuous exposure of the individual to the Viruddha food substances.
- As per Ayurveda literatures, Asatmya Ahara consumption is considered as Nidana or Anupashaya for any of the disease manifestation, it may even be responsible for the manifestation of Pandu disease.

Ca.I. – 1/10

Ca.Su. – 26/102,103

A.Hr.NI. – 1/7

- **Nishpava**¹² – consumption of beans
- **Masha**¹³ – consumption of black gram items
- **pinyaka** – consumption of *Pinyaka*
- **Tila Taila**¹⁴ **Nishevana** – consumption of *sesamum* oil prepared substances
- **Vidagdhe Anne Divaspapna, Vyayama and Maithuna**¹⁵ – sleeping during day time, performing exercise, performing sexual activity after consumption of *Vidaha* substances or during the digestion of food
- **Pratikarma Vaishamya**¹⁶ – improper diet and regimen during or after undergoing *Shodhana* line of treatment
- **Rtu Vaishamya** – improper diet and regimen related to a particular season
- **Vega Vidharana**¹⁷ – suppression of natural urges
- **Kama**¹⁸ – when the person is having excessive desire
- **Chinta**¹⁸ – when the person is having excess mental stress
- **Bhaya**¹⁸ – when the person is having excess fear

- **Krodha**¹⁸ – when the person is having excess anger
- **Shoka**¹⁸ – when the person is having excess grief
- **Vyavaya**¹⁸ – performing excess sexual activity
- **Madya**¹⁹ – excess consumption of intoxicating substances
- **Mrid**²⁰ – excess consumption of mud

Samprapti of Pandu

दोषाः पित्तप्रधानास्तु यस्य कुप्यन्ति धातुषु।
 शैथिल्यं तस्य धातूनां गौरवं चोपजायते॥४॥
 ततो वर्णवल्स्नेहा ये चान्येऽप्योजसो गुणाः।
 प्रजन्ति क्षयमत्यर्थं दोषदृष्यप्रदूषणात्॥५॥
 समुदीर्णो यदा पित्तं हृदये समवस्थितम्॥९॥
 वायुना बलिना क्षिप्तं संप्राप्य धमनीर्दश।
 प्रपत्रं केवलं देहं त्वयांसान्तरमाश्रितम्॥१०॥
 प्रदृष्य कफवाताकृत्यङ्ग्यांसानि करोति तत्।
 पाण्डुहारिद्रहरितान् वर्णान् बहुविधास्त्वचि॥११॥

Ca.Ci. – 16/4,5

Ca.Ci. – 16/9-11

- 12 कन्युष्णानि रुच्यानि वातलेप्महर कृतान्नेषूपयुज्यन्ते संस्कारार्थमनेकध

Su.Su. – 46/222

General qualities of vegetables as per Sushruta are Katu, Ushna Virya, palatable, pacifying Vata and Kapha Dosha. No description of qualities of Nishpava are mentioned. That is the reason, general qualities of vegetables should be taken.

- 13 वृष्यः परं बातहरः स्निग्धोष्णो मधुरो गुरुः। बल्यो बहुमलः पुंस्त्वं माषः शीघ्रं ददाति च॥

Ca.Su. – 26/24

Pandu is manifested in an individual due to the excess consumption of Masha. This occurs probably due to the increase of Snigdha quality, Ushna Virya, Guru quality. This in turn leads to the vitiation of Pitta Dosha and affliction of Rasa Dhatu.

- 14 तत्र तैलं हीण्यगीरवोपपन्नत्वाद्वातं जयति सततमभ्यस्यमानं चातो हि रौक्ष्यरीत्यापोपपो विरुद्धगुणो भवति, विरुद्धगुणसन्निपाते हि

Ca.Vi. – 1/14

भूयसाऽयमयजीयते, तस्मात्तैलं वातं जयति सत-

Due to qualities of Snigdha, Ushna qualities, there will be the vitiation of Pitta Dosha and because of the Snigdha and Gourava qualities, there may be the affliction of Rasa Dhatu during the manifestation of Pandu disease.

- 15 When an individual is undergoing sleeping during day time or exercise or sexual intercourse during the digestion of food substances, first of all, there will be reduced blood circulation to the alimentary canal which is an essential event happens. This is followed by deviation of blood supply to the skeletal muscles. This in turn leads to the improper digestion owing to the poor digestive secretion. In case of sleeping during day time, there will be excess digestive secretion secondary to parasympathetic nervous system stimulation occurs due to the sleeping during day time soon after the consumption of food.

- 16 Let us take the example of Snehana (a type of pre operative procedure, oleation therapy). When a person undergoes Snehana therapy, he has to consume hot water only, no other substances are consumed. This is especially helpful in preparing the patient for proper Shodhana line of treatment. While Snehana therapy, there will be therapeutic inflammation of the stomach, small intestine. Consumption of hot water helps in the mobility of adipose tissue (secondary to depletion of carbohydrate) along with pathogen, thereby leading to the emaciation of the body. If a person is consuming black gram items etc, this will provide adequate amount of carbohydrate to the body and the person will not have desired effect of mobilisation of fat.

- 17 Let us take the example of retention of urine because of suppression of urge of micturition. This may lead to poor renal blood supply resulting in the following events:

- Generalised oedema resulting from water and sodium retention
- Metabolic acidosis secondary to renal failure
- High concentration of no protein nitrogens, urea, creatinine and uric acid in particular

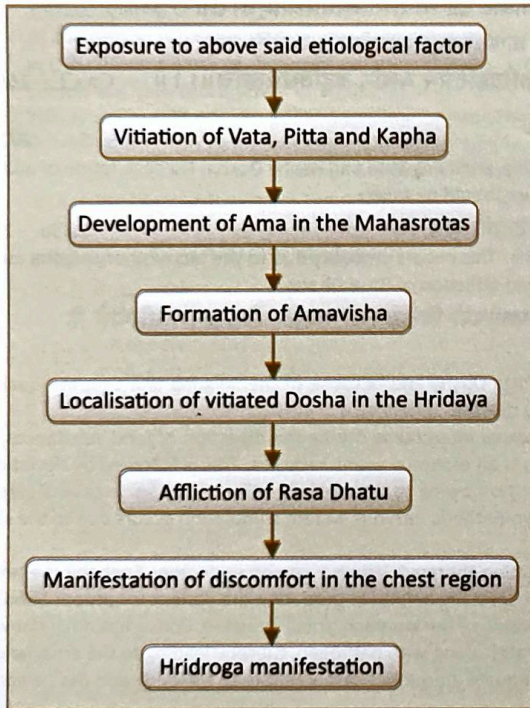
These may lead to the decreased production of erythropoietin which is an essential factor for the production of red blood corpuscles. All these kinds of activities are responsible for the activation of adrenal medullae which in turn responsible for the stimulation of sympathetic nervous system. This may lead to decreased alimentary secretions resulting the disturbance in the absorption of intrinsic factor, vitamin B12 and folic acid. Decreased renal arterial supply leads to the disturbance in the absorption of erythropoietin which leads to hampered blood formation.

- 19 Excess alcohol consumption will lead to the disturbed metabolism similar to starvation. This will lead to the poor absorption of the substances from the alimentary canal as the person is consuming less amount of food substances.

- Another hypothesis can be put forth. When an individual consumes more amount of alcohol, that may lead to the inflammation of stomach, which may in turn lead to the gastric damage, which in turn disutbe the absorb the vitamin B12 which is one of the essential factor for the formation of haemoglobin.

- 20 Excess consumption of mud is usually in the women who have undernutrition. Iron deficiency anemia is one of the commonest condition observed in the people who are consuming excess mud.

- Due to the consumption of *Nidana*, there will be vitiation of *Dosha*, *Pitta* in particular
- Vitiated *Dosha* then afflicts *Rakta Dhatu* as a result it brings laxity and heaviness in the *Dhatu*
- Heaviness of *Dhatu* manifest due to the disturbance of normal functioning of *Dhatu*. Patients suffering from *Pandu* lose their vitality, complexion, unctuousness and other properties of *Ojas* due to the morbidity of *Dosha* and *Dushya*.
- Aggravated *Pitta* expelled from *Hridaya* via ten channels, by powerful *Vata* circulates all over the body. Aggravated *Pitta* reaches the space between *Tvak* and *Mamsa* and brings about the abnormality in *Kapha*, *Vata*, *Rakta*, *Tvak* and *Mamsa* leading to the abnormal complexion like yellow, deep yellow and greenish discolouration.



- Aggravated *Pitta* is responsible for the less production of *Poshaka* (nutrient) portion from *Rasadhatu*, leading to decreased formation of *Rakta Dhatu* in the body.

Samprapti Ghataka

- Dosha:** *Pitta Pradhana Tridosha* (*Pitta* – *Sadhaka*; *Ranjaka* and *Bhrajaka* *Pitta*); *Vata* – *Vyana Vata*; *Kapha* – *Avalambaka*
- Dushya:** *Rasa*
- Agni:** *Jatharagni*, *Rasa Dhatvagni*
- Ama:** *Jatharagnijanya*
- Srotas:** *Rasavaha*, *Raktavaha Srotas*
- Srotodushti:** *Sanga* and *Vimargagamana*
- Udbhava sthana:** *Amashaya*
- Sanchara Sthana:** *Sarva Sharira*
- Vyakta Sthana:** *Tvak*
- Vyadhi Svabhava:** *Chrakari*
- Rogamarga:** *Bahya*
- Sadhysadhyata:** depending on the nature of vitiated *Dosha*, duration of illness

Purvarupa of Pandu

तस्य लिङ्गं भविष्यतः।

हृदयस्यन्दनं रौक्ष्यं स्वेदाभावः श्रमस्तथा॥12॥

Ca.Ci. – 16/12

त्वप्स्फोटनशीवनगात्रसादमुद्धक्षणाप्रेक्षणाकूटशोथाः।

विण्मूत्रपीतत्वमथाविपाको भविष्यतस्तस्य पुरःसराणि॥13॥

Ma.Ni. – Pandu/3

- Hridaya Spandana²¹:** sensation of palpitation of chest
- Roukshya²²:** dryness of the body
- Svedabhava²³:** absence of sweating
- Shrama²⁴:** exhaustion after doing little amount of activity
- Tvak Sphotana²⁵:** cracks in the skin
- Shtivana:** sputum production²⁶

G
K

- This is attributable to poor blood supply to the heart via coronary vessels. Thus, when a person starts doing the physical activity, he may have awareness of one's own heart beat. This is observed when an individual is having hypovolemia.
- This is attributable to either peripheral vaso constriction to shift the continuous blood supply to the core of the body i.e. brain and heart or mobilisation of fat owing to increased metabolic activity which can occur as a result of anaerobic metabolic activity.
- This is attributable to poor peripheral vaso constriction due to the shift of the blood from the skin to the core of the body i.e. brain and heart.
- This is attributable to decreased store of the adipose tissues from subcutaneous tissues, buttocks, breast and abdomen which occurs as a result of fat mobilisation secondary to lack of carbohydrate availability to the body.
- This is attributable to peripheral vaso constriction.
- This can be attributable to response of the human body to the respiratory allergens or pathogens or microorganisms entering through respiratory tract which may happen as a result of lowered immune status.
 - Lowered immunity may occur as a result of poor formation of WBC, which is formed due to the pluripotent haemopoietic stem cells (PHSC). These precursor cells are produced in red bone marrow present in the bones.
 - In anemia, aplastic anemia in particular, all cells RBCs, WBCs, platelets are produced in less quantities. That may be the reason for the susceptibility of infection.
 - Respiratory tract is the primary channel for the entry of most of the pathogens.

- Sluggish movement of the body parts²⁷
- Eating the mud²⁸
- Swelling around orbital region²⁹
- Yellowish discolouration of urine and fecal matter³⁰
- Impaired digestion in the body³¹

Samanya Lakshana of Pandu

संभूतेऽस्मिन् भवेत् सर्वः कर्णक्ष्वेडी हतानलः।

दुर्बलः सदनोऽद्विद् श्रमभ्रमनिपीडितः॥1 3॥

गात्रशूलज्वरश्चासगौरवा रुचिमात्रः।

सृदितैरिव गात्रै पीडितोन्मथितैरिव॥1 4॥

शूनाक्षिकूटो हरितः शीर्णलोमा हतप्रभः।

कोपनः शिशिरद्वेषी निद्रालुः छीवनोऽल्पवाक्॥1 5॥

पिण्डिकोद्वेष्टकं द्यूरुपादरुक्सदनानि च।

भवन्त्यारोहणायार्थविशेषश्चास्य वक्ष्यते॥1 6॥

Ca Ci. - 16/13-16

- Ringing in the ears³² e.g. tinnitus
- Loss of appetite³³
- Weakness of the body³⁴
- Sluggish movements of the body³⁵
- Aversion to take food substances³⁶
- Development of fatigue after doing the activities³⁷
- Giddiness³⁸
- Pricking type of pain all over the body³⁹
- Fever⁴⁰
- Breathing discomfort⁴¹
- Heaviness of the body parts⁴²
- Tastelessness⁴³

27 This is attributable to a poor blood circulation among small vessels of the body i.e. sludging of blood.

28 Excess consumption of mud is usually in the women who have undernutrition. Iron deficiency anemia is one of the commonest condition observed in the people who are consuming excess mud.

29 This can be attributable to pica, where the person is likely to eat mud. This is not a healthy practice, as this may lead to several infections of alimentary canal or helminthic manifestation.

30 This may be attributable to either hepatocellular jaundice or obstructive jaundice. In this situation, the person may have either poor protein synthesis or deficiency of other nutrients. When an individual is having increased RBC destruction, the person may have lemon yellow tinge discolouration of the skin and mucous membranes.

31 In case of impaired digestion in the stomach, the person may have decreased absorption of intrinsic factor whereas impaired digestion in the large intestine, there is a lack of absorption of vitamin B12, folic acid etc. Thus, impaired digestion may have impaired digestion.

32 This is attributable to shift of blood supply from the ears to the core of the body i.e. brain and heart. Thus, the person may have tinnitus if he is having anemia. This shift may lead to the reduction in the formation of perilymph and endolymph in the inner ears.

33 This is attributable to poor hydrochloric acid formation which further leads to the deficient absorption which further leads to the deficit in the absorption of intrinsic factor. When an individual is having disturbance in the large intestine, there will be increased secretion in the colon and further there will be decreased secretion of vitamin B12 and folic acid. These are the essential factors for the formation of blood. The person having loss of appetite for longer time, he may have loss of appetite.

34 This is attributable to poor storage of carbohydrate which in turn leading to the mobilisation of stored fat. This will lead to the emaciation of body and this may be the reason for the weakness of the body.

35 This may be attributable to hypovolemia. This may occur in case of haemolysis or in case of profuse haemorrhage etc. This may occur as a result of cardiac depression or increased capillary permeability or release of toxins by ischemic tissue or sludging of blood in small vessels.

36 This may be attributable to dilute hydrochloric acid which in turn leads to inactivation of pepsinogen and the person may have loss of appetite and this in turn leads to the aversion to take food substances. This may lead to deficit in the absorption of intrinsic factor and vitamin B12 and folic acid.

37 This can be attributable to hypovolemia. In this situation, there will be more blood circulation to the core of the body i.e. brain and heart and there will be less circulation to the skeletal muscles, thus, when an individual makes activity, he will have more fatigue which is a sequel of poor blood supply to the skeletal muscles.

38 This may be attributable to syncopal episodes which leads to poor cerebral circulation.

39 This can be attributable to generalised cellular degeneration. Following effects are observed in case of cellular degeneration.

1. Diminished active transport of sodium and potassium ions through the cell membrane. Due to this, sodium and chloride accumulate in the cells and potassium is lost from the cells leading to hydropic degeneration.
2. Mitochondrial activity in the liver cells and in many other tissues of the body, becomes severely depressed.
3. Lysosomes in the cells in widespread tissue areas begin to break open, with intracellular release of hydrolases that cause further intracellular degeneration.
4. Cellular metabolism of nutrients, such as glucose, eventually becomes greatly depressed in the last stages of shock.

The above said metabolic derangements lead to the manifestation of metabolic acidosis of the human body.

This may be responsible for the pricking type of pain all over the body.

40 This may be attributable to proliferation of microorganisms leading to progression of infection. This may occur because of deficient formation of PHSC (pluripotent haemopoietic stem cell) which is a precursor of RBC, WBC and platelets. Lack of formation of PHSC may be the reason for the frequent episode of fever in an individual.

41 This may be attributable to increased cardiac burden leading to tachycardia. Breathlessness can be observed either in sitting and recumbent posture unlike cardiac failure where initially the person will have breathlessness on exertion. On severity of cardiac disease, he will have orthopnoea and in fatal cardiac disease, he will have paroxysmal nocturnal dyspnoea and in severe form of cardiac dysfunction, he will have breathlessness during inspiration or expiration.

Cheyne Stokes respiration. Whereas in case of respiratory disease, he will have breathlessness during inspiration or expiration.

42 Refer the context of sluggish movements of the body

43 Refer the context of loss of appetite

- ❖ Patient feels as if his limbs are kneaded, pressed or churned
- ❖ Swelling in the periorbital region
- ❖ Greenish discolouration in the skin and body parts
- ❖ Loss of hairs from the body^{44,45}
- ❖ Lack of lustre⁴⁶
- ❖ Irritability⁴⁷
- ❖ Aversion to cold substances⁴⁸
- ❖ Excessive sleepishness⁴⁹
- ❖ Spitting⁵⁰
- ❖ Less speech⁵¹
- ❖ Muscle cramps in the calf region⁵²
- ❖ Pain the waist region⁵⁰
- ❖ Pain in the thigh region⁵⁰
- ❖ Pain in the lower limbs⁵⁰
- ❖ Breathlessness on exertion⁵³

2. Pittaja
3. Kaphaja
4. Sannipataja
5. Mrid Bhakshanajanya

As per Sushruta – 4 types

1. Vataja
2. Pittaja
3. Kaphaja
4. Sannipataja

Vataja Pandu Lakshana

आहारैरुपचारैश्च वातलैः कुपितोऽनिलः।

जनयेत्कृष्णपाण्डुत्वं तथा रुक्षारुणाङ्गताम्॥ 1 7 ॥

अङ्गमर्दं रुजं तोदं कम्पं पाश्चशिरोरुजम्।

वर्चःशोषास्यवैरस्यशोफानाहवलक्षयान्॥ 1 8 ॥

Ca.Ci. – 16/17-18

- ❖ Blackish discolouration
- ❖ Yellowish white discolouration
- ❖ Roughness of the body parts
- ❖ Redness of the body parts

Sankhya Samprapti of Pandu

As per Charaka – 5 types

1. Vataja

- 44 This is attributable to poor peripheral arterial circulation which results in thinning of the skin and shininess of the skin. This is attributable to peripheral vaso constriction resulting in deviation of blood supply to the core of the body i.e. brain and heart.
- 45 This is attributable to poor peripheral circulation which leads to thinning of the skin, loss of the hairs from the body
- 46 This is attributable to poor peripheral circulation as it will lead to cracks of the skin or dryness of the skin and there will be less moisture of the skin in due course of time.
- 47 This is attributable to poor cerebral circulation as a neurological dysfunction.
- 48 This is attributable to peripheral vaso constriction where in the person will have feeling of coldness. As a result, the person will often have aversion to cold exposure.
The human beings can able to perceive different gradations of cold and heat, from freezing cold to cold to cool to indifferent to warm to hot to burning hot sensations. Thermal gradations are discriminated by at least 3 types of sensory receptors: cold receptors, warmth receptors and pain receptors.
The cold and warm receptors are located immediately under the skin at discretely separated spots. In most parts of the body, there are 3 to 10 times as many cold spots as warm spots. The number in different areas of the body varies from 15 to 25 cold spots per square centimeter in the lips to 3 to 5 cold spots per square centimeter in the finger to less than 1 cold spot per square centimeter in some broad surface areas of the trunk.
When a cold receptor is suddenly subjected to an abrupt fall in the temperature, it becomes strongly stimulated at first, but this stimulation fades rapidly during the first few seconds and progressively more slowly during the next 30 minutes or more.
Cold and warm receptors are stimulated by the changes in their metabolic rates and these changes result from alteration in the rate of intracellular chemical reactions more than 10° C. In other words, thermal detection probably results not from direct physical effects of heat or cold on the nerve endings, but from chemical stimulation of nerve endings as modified by temperature.
- 49 Excessive sleepishness can be attributable to poor cerebral circulation as a result of syncopal attack. It occurs in severe variety of anemia.
- 50 When an individual is exposed to multiple aetiological factors, three parts of the body are exposed – gastrointestinal tract, upper respiratory passages and skin. Skin is considered to be the most stable organ as far as the appearance of clinical symptoms in relation to exposure to aetiological factors are concerned.
Owing to the poor formation of pluripotent haemopoietic stem cell (PHSC), RBCs, WBCs and platelet formation become reduced, thereby the person may be vulnerable to the respiratory compromise which is leading to the formation of sputum during the course of the disease.
- 51 As per Ayurveda fundamentals, Karma (functions) of human body is of three categories – Kayika (physical), Vachika (speech related) and Manasika (mental). Whereas as per western medical understanding, functions are either physical or mental. For any activity to happen smoothly, it requires certain amount of energy and in the person suffering from Pandu, he will have Nissara (lack of nutrition). Kayika, Vachika and Manasika functions get disturbed in the person who is suffering from Pandu. In olden days, speech related function was related to study of Vedas, treasure of knowledge. It is a custom that Vedas are not written by anybody, the verses present in Veda are recited and it requires huge amount of energy for proper pronunciation. In Pandu patient, speech becomes difficult.
- 52 These clinical features are attributable to anaerobic metabolic activities leading to the excess concentration of lactic acid manifesting the clinical condition called metabolic acidosis.
- 53 This is attributable to tachycardia develops after performing physical exercise and during this activity, there will be poor blood supply to myocardium leading to breathlessness on exertion.

- General malaise
- Discomfort
- Pricking type of pain
- Tremors
- Pain in the sides of the chest
- Pain in the head region
- Dryness of stools (very hard stools)
- Impaired taste perception
- Swelling
- Distension of the abdomen
- Loss of strength

Pittaja Pandu Lakshana

- पित्तलस्याचितं पित्तं यथोक्तः स्वैः प्रकोपणैः।
 दूषयित्वा तु रक्तादीन् पाण्डुरोगाय कल्पते॥19॥
 स पीतो हरिताभो वा ज्वरदाहसमन्वितः।
 तृष्णामूर्च्छापिपासार्तः पीतमूत्रशकृन्नरः॥20॥
 स्वेदनः शीतकामश्च न चान्नमभिनन्दति।
 कटुकास्यो न चास्योष्णमुपशेतेऽम्लमेव च॥21॥
 उद्गारोऽम्लो विदाहश्च विदग्धेऽन्नेऽस्य जायते।
 दौर्गन्ध्यं भिन्नवर्चस्त्वं दौर्बल्यं तम एव च॥22॥ Ca.Ci. – 16/19-22
- Yellowish discolouration of the body parts
 - Greenish discolouration of the body parts
 - Fever
 - Burning sensation
 - Thirst
 - Altered state of consciousness
 - Morbid thirst
 - Yellowish discolouration of urine
 - Yellowish discolouration of faecal matter
 - Excessive sweating
 - Affinity or liking towards cold substances
 - Aversion to take food substances
 - Pungent taste in the mouth
 - Patient will be uncomfortable to take hot and sour substances
 - Sour eructations
 - Burning sensation during the digestion of food
 - Foul smell from the body
 - Unformed stool or diarrhoea
 - Debility
 - Sensation of darkness in front of the eyes

Kaphaja Pandu Lakshana

विबुद्धः श्लेष्मलैः लेष्मा पाण्डुरोगं स पूर्ववत्।

करोति गौरवं तन्द्रां छर्दिं श्वेतावभासताम्॥23॥
 प्रसेकं लोमहर्षं च सादं मूर्च्छां भ्रमं क्लमम्।
 श्वासं कासं तथाऽऽलस्य मरुचि वाक्स्वरग्रहम्॥24॥
 शुक्लमूत्राक्षिवर्चस्त्वं कटुरुक्षोणका मताम्।
 श्वयथुं मधुरास्यत्वमिति पाण्डामयः कफात्॥25॥ Ca.Ci. – 16/23-25

- Heaviness of the body parts
- Drowsiness
- Vomiting
- Whitish discolouration of the body parts
- Excess salivation
- Horripulation
- Heaviness or general malaise
- Altered state of consciousness
- Giddiness
- Fatigue without doing any activities
- Breathing discomfort
- Cough
- Laziness
- Tastelessness
- Loss of speech and voice
- Whitish discolouration of the eyes
- Whitish discolouration of the urine
- Whitish discolouration of faecal matter
- Liking to take pungent substances
- Liking to take dry substances
- Liking to take hot substances
- Swelling or oedema
- Sweetish taste in the mouth

Sannipataja Pandu

Mrid Bhakshanajanya Pandu Lakshana:

मृत्तिकादनशीलस्य कुप्यत्यन्यतमो मलः।
 कषाया मारुतं, पित्तमूषरा, मधुरा कफम्॥27॥
 कोपयेन्मृद्रसादींश्च रौक्ष्याद्भुक्तं विरुक्षयेत्।
 पूरयत्यविषचैव स्रोतांसि निरुणद्धि च॥28॥
 इन्द्रियाणां बलं हत्वा तेजो वीयजसी तथा।
 पाण्डुरोगं करोत्याशु बलवर्णाग्निनाशनम्॥29॥
 शूनगण्डाक्षिकूटभूः शूनपान्नाभिमेहनः।
 क्रिमिकोष्ठोऽतिसार्येत मलं सास्क कफान्वितम्॥ Ca.Ci. – 16/27-30

- One who consumes mud (clay) causes aggravation of either Vata, Pitta or Kapha. Consumption of mud having astringent taste vitiates Vata, consumption of mud having alkaline nature will vitiate Pitta and consumption of mud having sweet taste will vitiate Kapha Dosha.

- ❖ Due to the ununctuousness, there will be vitiation of *Rasa* etc due to the consumption of mud.
- ❖ Poor functioning of *Indriya*
- ❖ Loss of strength
- ❖ Loss of complexion
- ❖ Loss of digestive function
- ❖ Swelling of the temporal region
- ❖ Swelling of the periorbital region
- ❖ Swelling of eyes
- ❖ Swelling of lower limbs
- ❖ Swelling of periumbilical region
- ❖ Swelling of genital organs
- ❖ Presence of *Krimi* in the *Mahasrotas*
- ❖ *Diarrhoea* associated with mucus and blood

Asadhya Lakshana of Pandu

पाण्डुरोगश्चिरोत्पन्नः खरीभूतो न सिध्यति।

कालप्रकर्षाच्छूनो ना यश्च पीतानि पश्यति॥3 1

वाल्पविकं सकफं हरितं योऽतिसार्यते।

दीनः श्वेतातिदिग्धाङ्गच्छर्दिमूर्च्छातपादितः॥3 2 ॥

स नास्त्यक्षयाद्यश्च पाण्डुः श्वेतत्वमाप्नुयात्।

इति पञ्चविधस्योक्तं पाण्डुरोगस्य लक्षणम्॥3 3 ॥ Ca.Ci. - 16/31-33

पाण्डुरोगश्चिरोत्पन्नः खरीभूतो न सिध्यति।

कालप्रकर्षानां यो वा पीतानि पश्यति॥1 2 ॥

बद्धाल्पविट् सहरितं सकफं योऽतिसार्यते।

दीनः श्वेतातिदिग्धाङ्गच्छर्दिमूर्च्छादितः॥1 3 ॥

स नास्त्यसृक्क्षयाद्यश्च पाण्डुः श्वेतत्वमाप्नुयात्।

पाण्डुदन्तनखो यस्तु पाण्डुनेत्रश्च यो भवेत्।

पाण्डुसंघातदर्शी च पाण्डुरोगी विनश्यति॥1 4 ॥

अन्तेषु शूनं परिहीणमध्यं म्लानं तथाऽन्तेषु च मध्यशूनम्।

गुदे च शेषस्यथ मुष्कयोश्च शूनं प्रताम्यन्तमसंशकल्पम्।

विवर्जयेत्याण्डुकिं यशोऽथ तथाऽतिसारज्वरपीडितं च॥1 5 ॥

Ma.Ni. - Pandu/ 12-15

- ❖ The person suffering from *Pandu* for longer duration will be associated with **excessive roughness**⁵⁴
- ❖ Due to the chronicity, the person will be presented with **oedema**⁵⁵
- ❖ Patient sees the **objects as yellow**⁵⁶
- ❖ Constipation or very little amount of **faecal matter** will be formed⁵⁷
- ❖ **Diarrhoea** with greenish discolouration
- ❖ **Diarrhoea** with whitish discolouration or with mucus
- ❖ **Dull look**⁵⁸
- ❖ Excessive paleness as if he is **smeared**⁵⁶
- ❖ Excessive vomiting faint and excess thirst
- ❖ Whitish discolouration of the body parts due to **deficiency of the blood**⁵⁹
- ❖ **Swelling in the extremities**⁶⁰ and **emaciation in the trunk**⁶¹
- ❖ Swelling in both **extremities, trunk, anal region, genital organs and scrotum**⁶²
- ❖ **Frequent fainting**⁶³
- ❖ **Altered consciousness**⁶¹
- ❖ Associated with **diarrhoea**⁶⁴ or **fever**⁶⁵

G
K

- 54 This is attributable to peripheral vaso constriction which may persist for longer duration. In this situation, the person may have shift of the blood supply to the core of the body, brain and heart in particular which in turn may lead to roughness of the skin.
- 55 Oedema in the patient of anemia can be attributable to either hypoproteinemia (less concentration of proteins in the intravascular compartment or increased concentration of proteins in the interstitial compartment) or less concentration of haemoglobin in the blood which itself is a protein. This can also happen because of poor hepatic function or poor protein absorption from the alimentary canal.
- 56 This clinical condition can be observed in case of increased haemolysis or in case of hepatocellular dysmetabolism.
- 57 This can be attributable to poor circulation in the alimentary canal where shift of the circulation to the core of the body can be observed leading to poor gastrointestinal secretions leading to the very hard faecal matter. Thus, the person may have the clinical presentation of constipation.
- 58 These features are attributable to poor peripheral circulation leading to the pallor which can be further seen in eyes, tongue, palm, nails and even in the skin.
- 59 This can be attributable to poor peripheral circulation and increased core circulation, brain and heart in particular. This can be observed in severe forms of anemia.
- 60 This can be attributable to poor circulatory failure where we can observe hypovolemic shock, where sludging of the blood in very small blood vessels can be present.
- 61 This can be attributable to poor cellular nutrition which initially occurs as carbohydrate depletion which in turn leads to the rapid depletion of the accumulated fat in the subcutaneous tissues in buttocks, breasts and abdomen in particular.
- 62 This can be attributable to excess hypoproteinemia which can lead to reduced oncotic pressure in the capillaries which in turn leads excess accumulation of the fluid in the interstitial compartment.
- 63 These are attributable to poor cerebral circulation and this may be observed in severe form of anemia.
- 64 This can be attributable to non absorption of substances formed in the large intestine because of bacterial activity which are vitamin K, vitamin B12, thiamine, riboflavin etc. Among these substances, vitamin B12 and thiamine are directly responsible for the the production of RBCs.
- 65 As far as genesis of blood cells are concerned, red blood corpuscles (RBCs), white blood corpuscles (WBCs) and platelets are formed by one maternal cell called pluripotent haemopoietic stem cell (PHSC). When the person is suffering from fever for longer duration, this will lead to the increased formation of WBCs and this will lead to decreased formation of RBCs and this mechanism may lead to the manifestation of

Upadrava of Pandua

उपद्रवास्तेष्वरुचिः पिपासा
छर्दिर्ज्वरो मूरुजाऽग्निसादः।
शोफस्तथा कण्ठगतोऽबलत्वं मूर्च्छा क्लमो हृद्यवपीडनं च॥ 3 ॥

Su.Ut. – 44/13

- ❖ Tastelessness
- ❖ Thirst
- ❖ Vomiting⁶⁶
- ❖ Fever⁶⁷

- ❖ Headache⁶⁸
- ❖ Loss of appetite
- ❖ Swelling in the throat region⁶⁹
- ❖ Loss of strength⁷⁰
- ❖ Altered state of consciousness⁷¹
- ❖ Fatigue without doing any activities⁷²
- ❖ Discomfort in the chest region⁷³

Key summary of the chapter

पाण्डुना वक्ष्यमाण हरितादि वर्णोभ्यः प्रधानेन वर्णोनोपलक्षितो रोगः पाण्डुरोगः॥

Chakra, Ca.Ci. – 16/1,2

- ❖ Panduroga is a disease which is characterised by abnormal discolouration of the skin, abnormal texture of the skin. This may be yellowish discolouration or whitish discolouration or yellowish discolouration. Abnormal texture of the skin may be in the form of cracks of the skin or rough skin can be seen in the patient who are suffering from discolouration.
- ❖ Pandu is a Rasavaha Srotovikara where Pitta Dosha dominance is observed, Rasa Dhatu involvement is identified throughout the disease process with Rasavaha Sroto involvement and Sanga and Vimargagamana is the type of Srotodushti,
- ❖ As far as Sankhya Samprapti of Pandu is concerned, 5 types of Pandu are explained. They are Vataja, Pittaja, Kaphaja, Sannipataja and Mrid Bhakshanaja Pandu. Prime Lakshana observed in Mrid Bhakshananya Pandu is character is Krimikoshtha.
- ❖ Any clinical condition which interferes in the transformation of nutrients which can be utilised by the human or any condition associated with profuse bleeding or any condition associated chronic infection are the probable causes of manifestation of anemia.

Previous questions asked in the university examinations

Long essay questions

- ❖ Define Pandu. Explain Nidana, Samprapti, Purvarupa, Rupa, Asadhya Lakshana of Pandu
- ❖ Define Pandu, explain Nidana Panchaka of Pandu
- ❖ Explain Nidana Panchaka of Pandu

Short essay questions

- ❖ Write the Pratyatma Lakshana of Pandu
- ❖ Write the Samprapti and Samprapti Ghataka of Pandu
- ❖ Explain Krimi Koshta

anemia.

- 66 While analysing vomiting in the patient of anemia, we have to understand two pathological concepts – first, there will be a possibility of haematemesis or malena. Secondly, we have to understand the possibility of reduction of morrhage in the alimentary canal leading to either haematemesis or malena. When there is deficiency of intrinsic factor, there is a possibility of more serious consequences such as nutrition deficiency anemia etc.
- 67 This can be attributable to the poor WBC function which occurs as a sequel of poor formation of pluripotent haemopoietic stem cell (PHSC) which results in rapid manifestation of infection in the individuals who are having poor PHSC.
- 68 This can be attributable to either respiratory infection or allergic respiratory manifestation which leads to the paranasal sinus inflammation or infection. This usually presents with headache.
- 69 Whenever, the person is suffering from chronic anemia, one of the mechanism happens is formation of webs and rings in the oesophagus. Swelling in the throat refers to the appearance of ring in upper oesophageal junction. In the western medicine, this is termed as Plummer Vinson syndrome.
- 70 This can be attributable to poor peripheral circulation which in turn may lead to hyponatremia, hypokalemia, hypocalcemia and dehydration. As every activity involving either skeletal muscles, smooth muscles, cardiac muscles and every nerve conduction requires the above mentioned electrolytes and water, loss or poor strength is inevitable in the patient of Pandu (anemia).
- 71 This is attributable to poor cerebral circulation. In this occasion, the person may have shift of blood circulation to the skeletal muscles. In this situation, the person may have shift of blood circulation to the skeletal muscles.
- 72 This can be attributable to poor blood supply to the skeletal muscles. In this occasion, the person may have shift of blood circulation to the skeletal muscles. In this situation, the person may have shift of blood circulation to the skeletal muscles.
- 73 This can be attributable to poor blood supply to the skeletal muscles. In this occasion, the person may have shift of blood circulation to the skeletal muscles. In this situation, the person may have shift of blood circulation to the skeletal muscles.

- ⊗ Explain Mrid Bhakshanajanya Pandu
- ⊗ Enumerate the types of Pandu

MCQs

1. Krimi Koshtha is observed in which of the following type of Pandu?
 - a. Kaphaja Pandu;
 - b. Vataja Pandu;
 - c. Mrid Bhakshanajya Pandu;
 - d. All of the above

Answer: (c)

2. Mrid Bhakshana in the causation of Pandu. Here, the term Mrid Bhakshana refers to
 - a. Jnapaka Hetu;
 - b. Pradhanika Hetu;
 - c. Vyadhi Hetu;
 - d. Vyanjaka Hetu

Answer: (c)

3. Sankhya Samprapti of Pandu is
 - a. 4;
 - b. 3;
 - c. 5;
 - d. 2

Answer: (c)

4. Pandu isSrotogata Vikara.
 - a. Mamsavaha;
 - b. Rasavaha;
 - c. Raktavaha;
 - d. Medovaha

Answer: (b)

5. Sankhya Samprapti of Pandu as per Sushruta is
 - a. 3;
 - b. 4;
 - c. 5;
 - d. 7

Answer: (b)

6. Hrid drava is the Purvarupa of
 - a. Tamaka Shvasa;
 - b. Raktapitta;
 - c. Pandu;
 - d. Amavata

Answer: (c)

7. Pradhana Dosha involved in the manifestation of Pandu is
 - a. Vata;
 - b. Pitta;

- c. Kapha;
- d. Vata and Kapha

Answer: (b)

8. The term 'Anteshu Shunam Parihina Madhyam' is associated with
 - a. Shotha;
 - b. Raktapitta;
 - c. Pandu;
 - d. Gulma

Answer: (c)

9. The lifespan of senescent RBC is usually
 - a. 100 days;
 - b. 80 days;
 - c. 150 days;
 - d. 120 days

Answer: (d)

10. Sankhya Samprapti of Pandu as per Harita is
 - a. 4;
 - b. 5;
 - c. 6;
 - d. 7

Answer: (c)

Chapter resources

- ⊗ <https://arunachalagrace.blogspot.com/2010/02/>
- ⊗ Charaka Samhita, Chikitsa Sthana – 16th chapter: Pandu Chikitsa Adhyaya
- ⊗ Sushruta Samhita Uttara Tantra – 44th chapter: Panduroga Pratishedha Adhyaya
- ⊗ Ashtanga Hridaya Nidana Sthana – 13th chapter: Panduroga Shopha Visarpa Nidana Adhyaya
- ⊗ Ashtanga Sangraha Nidana Sthana – 13th chapter: Pandu Shopha Visarpa Nidana Adhyaya
- ⊗ Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- ⊗ The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- ⊗ Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- ⊗ Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- ⊗ Textbook of Medical physiology by Arthur C Guyton
